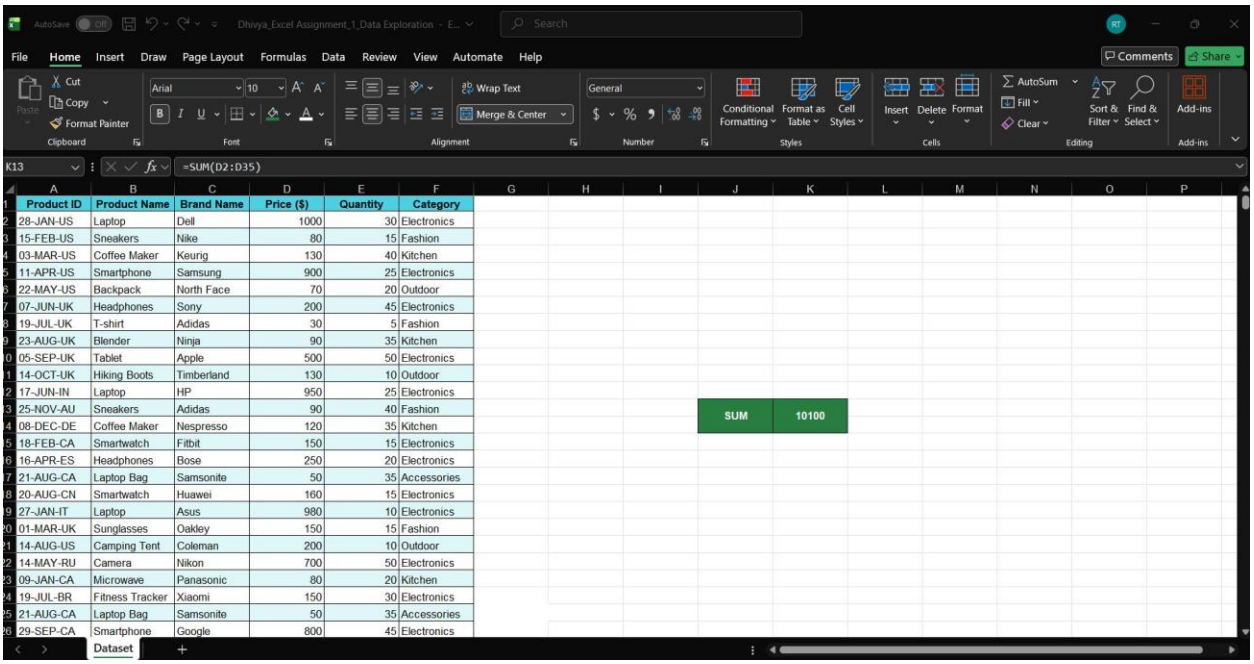
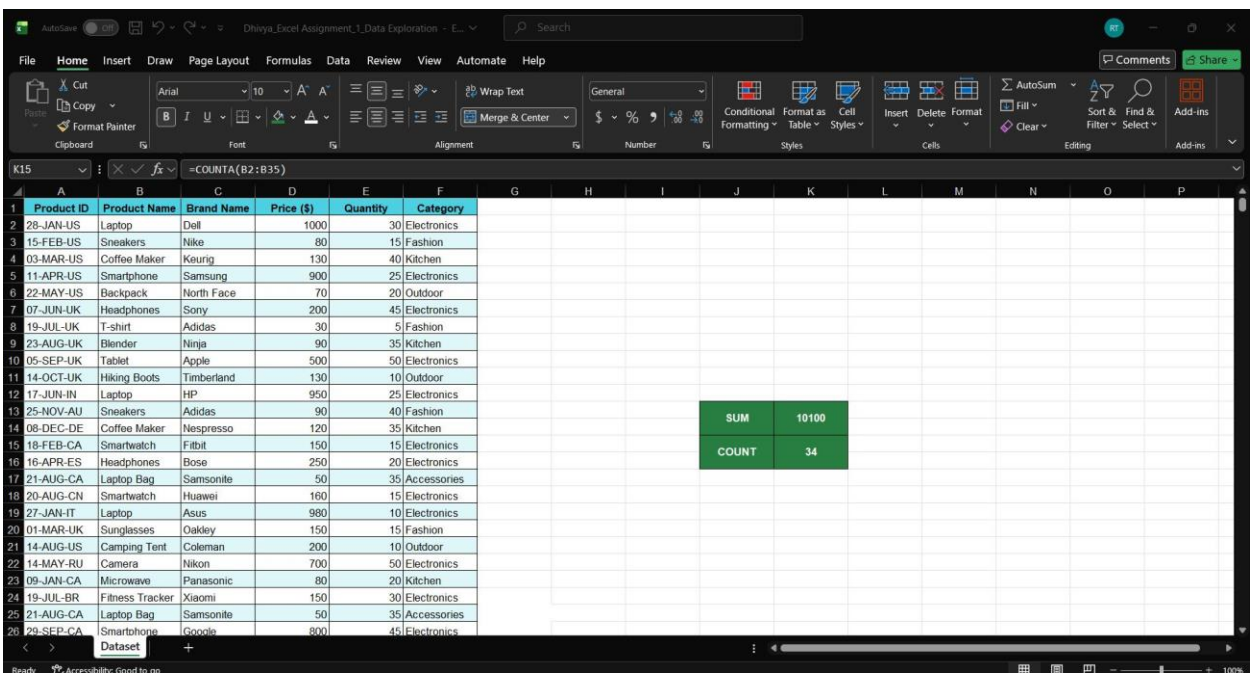
Assignment 1 Data Exploration

1. Sum, Count, and Average

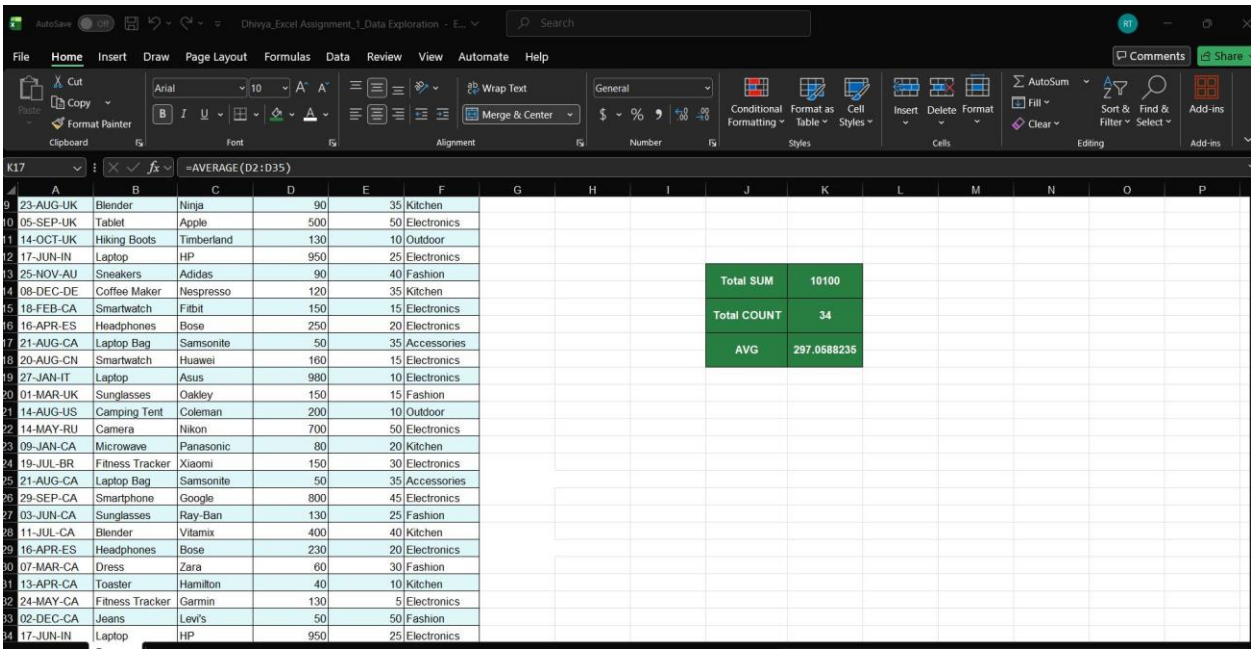
SUM FUNCTION =SUM(D2:D35)



COUNT FUNCTION =COUNTA(B2:B35)



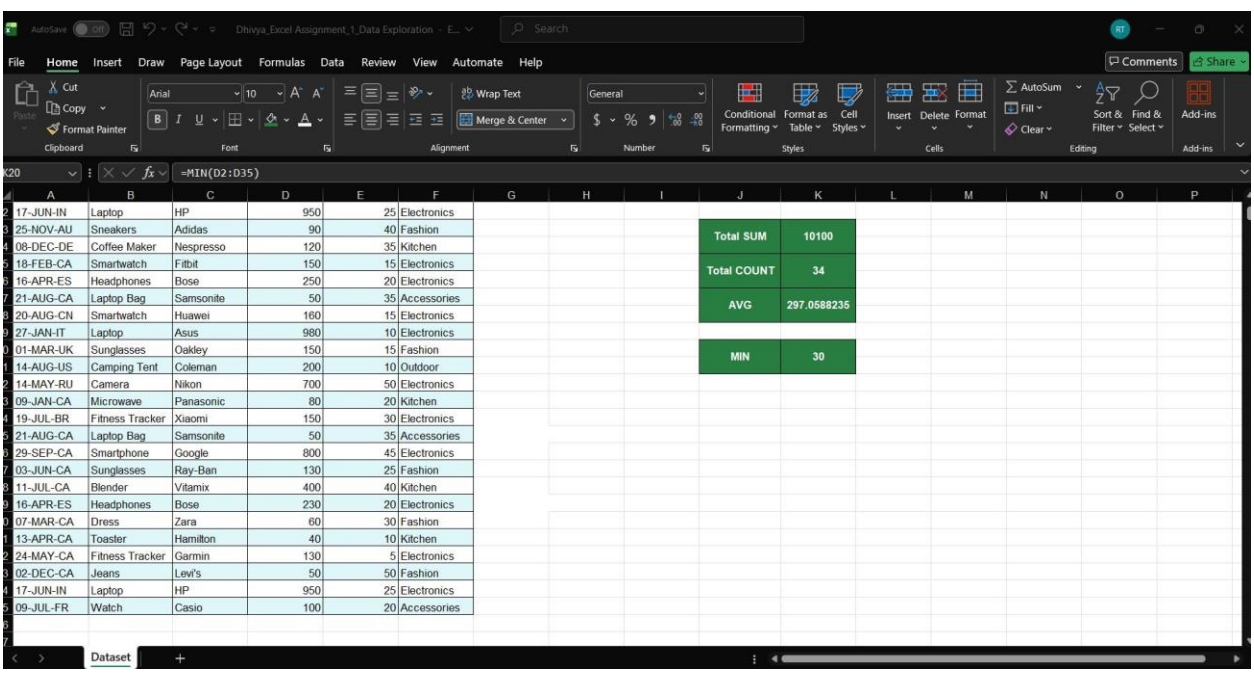
AVG FUNCTION =AVERAGE(D2:D35)



23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	Headphones	Bose	230	20	Electronics
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics

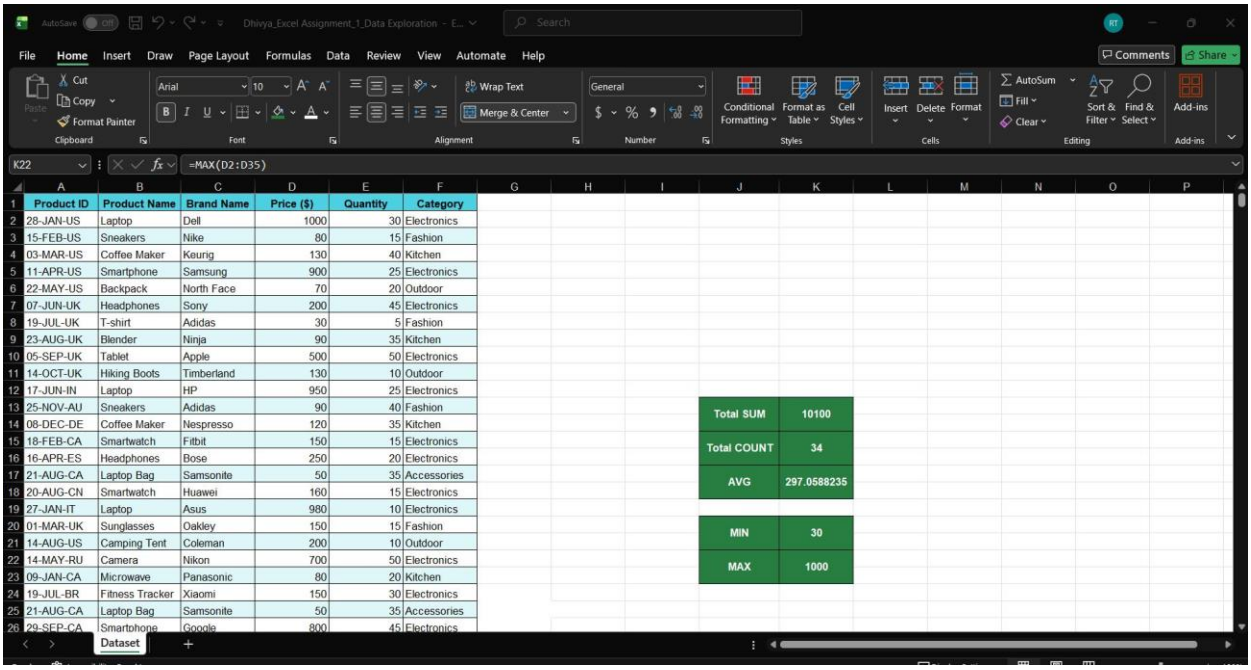
2. Minimum and Maximum Values

MIN FUNCTION =MIN(D2:D35)



17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics
03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion
11-JUL-CA	Blender	Vitamix	400	40	Kitchen
16-APR-ES	Headphones	Bose	230	20	Electronics
07-MAR-CA	Dress	Zara	60	30	Fashion
13-APR-CA	Toaster	Hamilton	40	10	Kitchen
24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics
02-DEC-CA	Jeans	Levi's	50	50	Fashion
17-JUN-IN	Laptop	HP	950	25	Electronics
09-JUL-FR	Watch	Casio	100	20	Accessories

MAX FUNCTION =MAX(D2:D35)

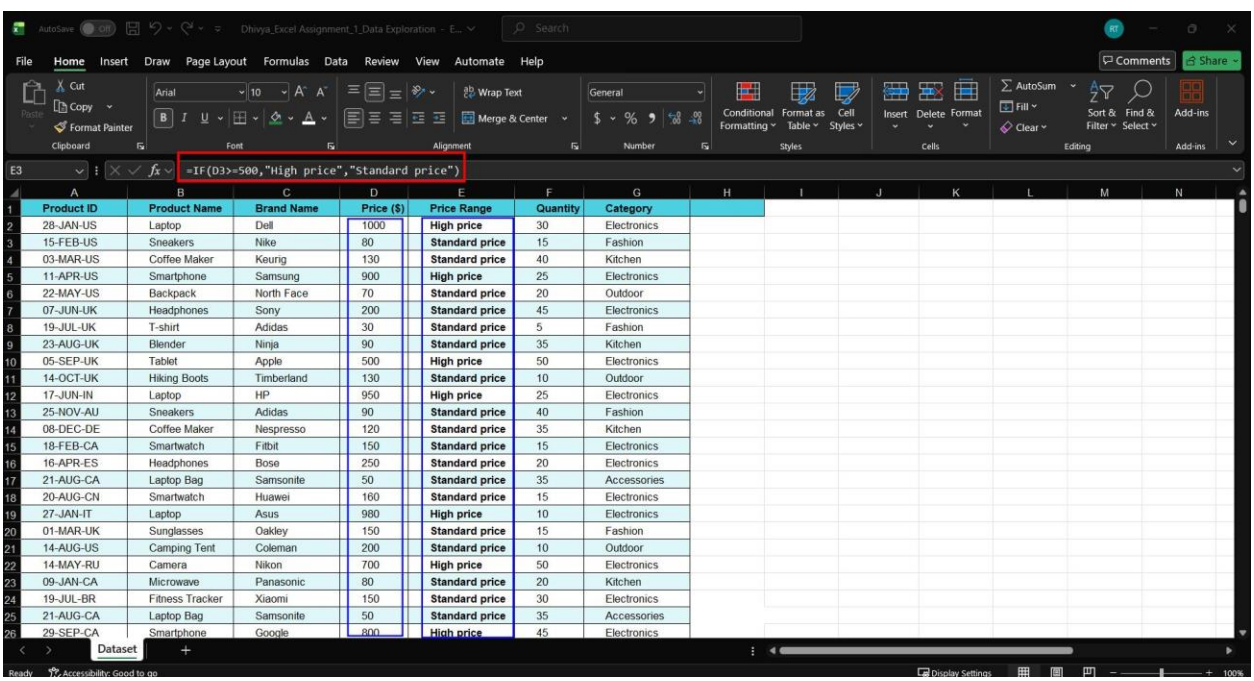


Product ID	Product Name	Brand Name	Price ($)	Quantity	Category
28-JAN-US	Laptop	Dell	1000	30	Electronics
15-FEB-US	Sneakers	Nike	80	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen
11-APR-US	Smartphone	Samsung	900	25	Electronics
22-MAY-US	Backpack	North Face	70	20	Outdoor
07-JUN-UK	Headphones	Sony	200	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	5	Fashion
23-AUG-UK	Blender	Ninja	90	35	Kitchen
05-SEP-UK	Tablet	Apple	500	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor
17-JUN-IN	Laptop	HP	950	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics
16-APR-ES	Headphones	Bose	250	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	15	Electronics
27-JAN-IT	Laptop	Asus	980	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	10	Outdoor
14-MAY-RU	Camera	Nikon	700	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories
29-SEP-CA	Smartphone	Google	800	45	Electronics

Total SUM	10100
Total COUNT	34
AVG	297.0588235
MIN	30
MAX	1000

3. Logical Function – IF

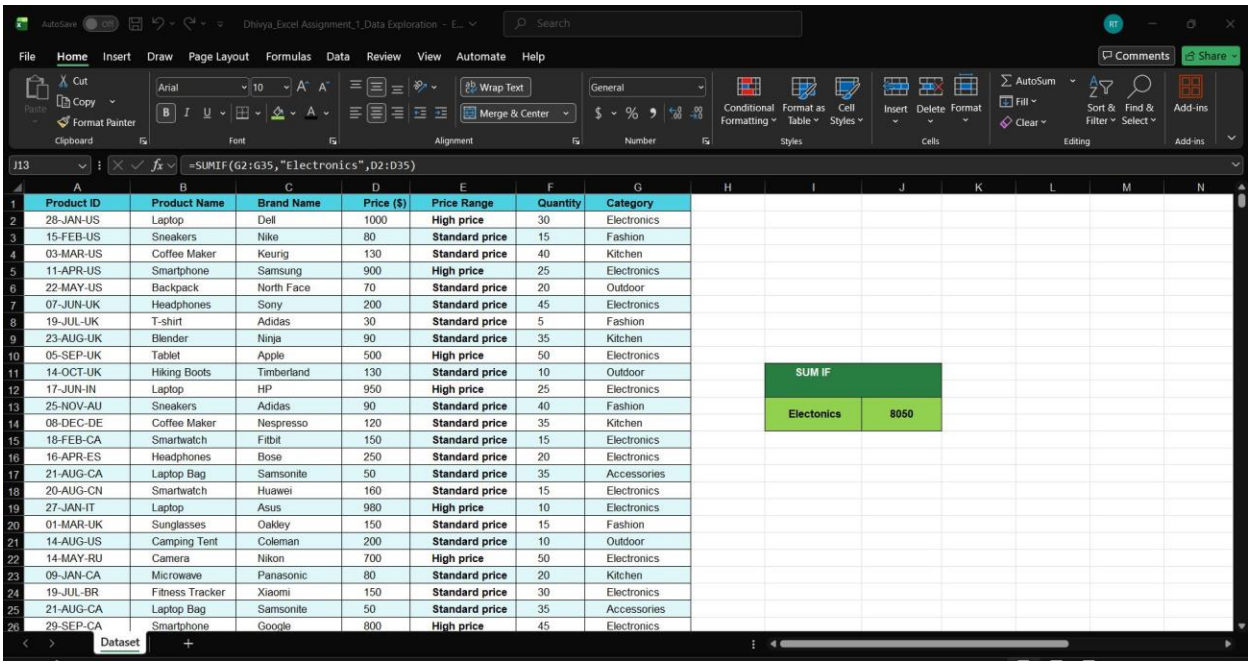
IF FUNCTION =IF(D2>=500,"High price","Standard price")



Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category
28-JAN-US	Laptop	Dell	1000	High price	30	Electronics
15-FEB-US	Sneakers	Nike	80	Standard price	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	Standard price	40	Kitchen
11-APR-US	Smartphone	Samsung	900	High price	25	Electronics
22-MAY-US	Backpack	North Face	70	Standard price	20	Outdoor
07-JUN-UK	Headphones	Sony	200	Standard price	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	Standard price	5	Fashion
23-AUG-UK	Blender	Ninja	90	Standard price	35	Kitchen
05-SEP-UK	Tablet	Apple	500	High price	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	Standard price	10	Outdoor
17-JUN-IN	Laptop	HP	950	High price	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	Standard price	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	Standard price	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	Standard price	15	Electronics
16-APR-ES	Headphones	Bose	250	Standard price	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	Standard price	15	Electronics
27-JAN-IT	Laptop	Asus	980	High price	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	Standard price	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	Standard price	10	Outdoor
14-MAY-RU	Camera	Nikon	700	High price	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	Standard price	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard price	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories
29-SEP-CA	Smartphone	Google	800	High price	45	Electronics

4. Conditional Functions – SUMIF and COUNTIF

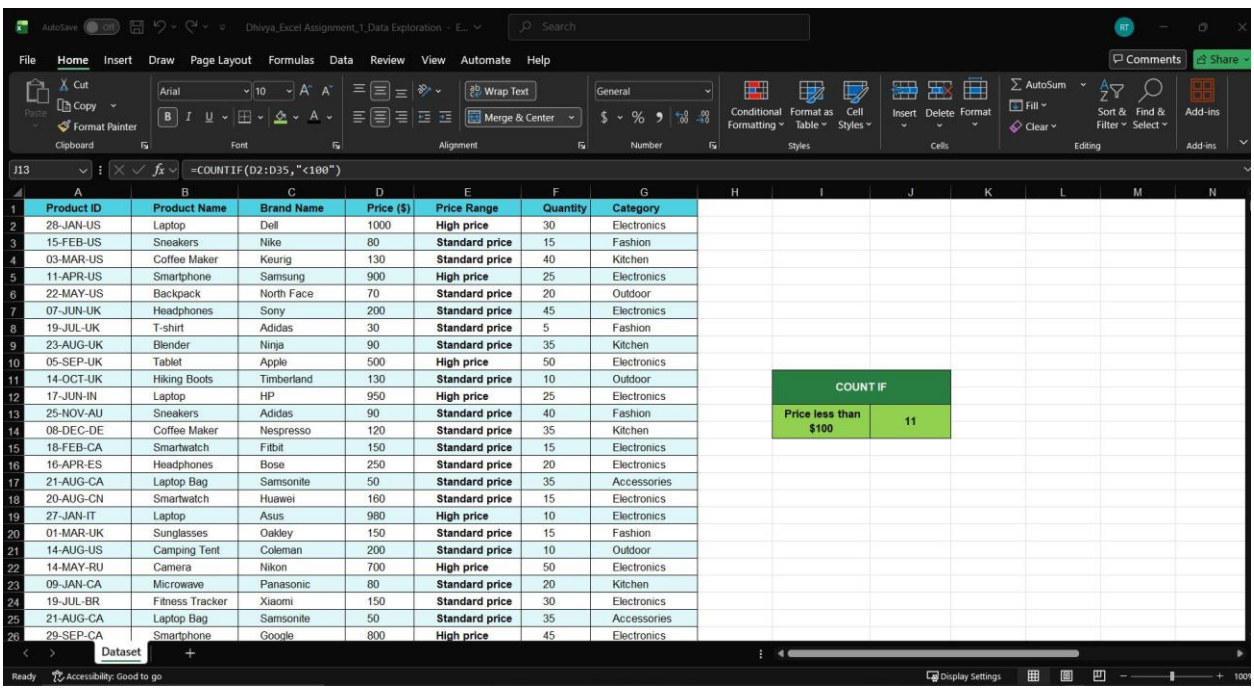
SUM IF FUNCTION =SUMIF(G2:G35, "Electronics", D2:D35)



Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category
28-JAN-US	Laptop	Dell	1000	High price	30	Electronics
15-FEB-US	Sneakers	Nike	80	Standard price	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	Standard price	40	Kitchen
11-APR-US	Smartphone	Samsung	900	High price	25	Electronics
22-MAY-US	Backpack	North Face	70	Standard price	20	Outdoor
07-JUN-UK	Headphones	Sony	200	Standard price	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	Standard price	5	Fashion
23-AUG-UK	Blender	Ninja	90	Standard price	35	Kitchen
05-SEP-UK	Tablet	Apple	500	High price	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	Standard price	10	Outdoor
17-JUN-IN	Laptop	HP	950	High price	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	Standard price	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	Standard price	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	Standard price	15	Electronics
16-APR-ES	Headphones	Bose	250	Standard price	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	Standard price	15	Electronics
27-JAN-IT	Laptop	Asus	980	High price	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	Standard price	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	Standard price	10	Outdoor
14-MAY-RU	Camera	Nikon	700	High price	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	Standard price	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard price	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories
29-SEP-CA	Smartphone	Google	800	High price	45	Electronics

SUM IF	
Electonics	8050

COUNT IF FUNCTION =COUNTIF(D2:D35,"<100")

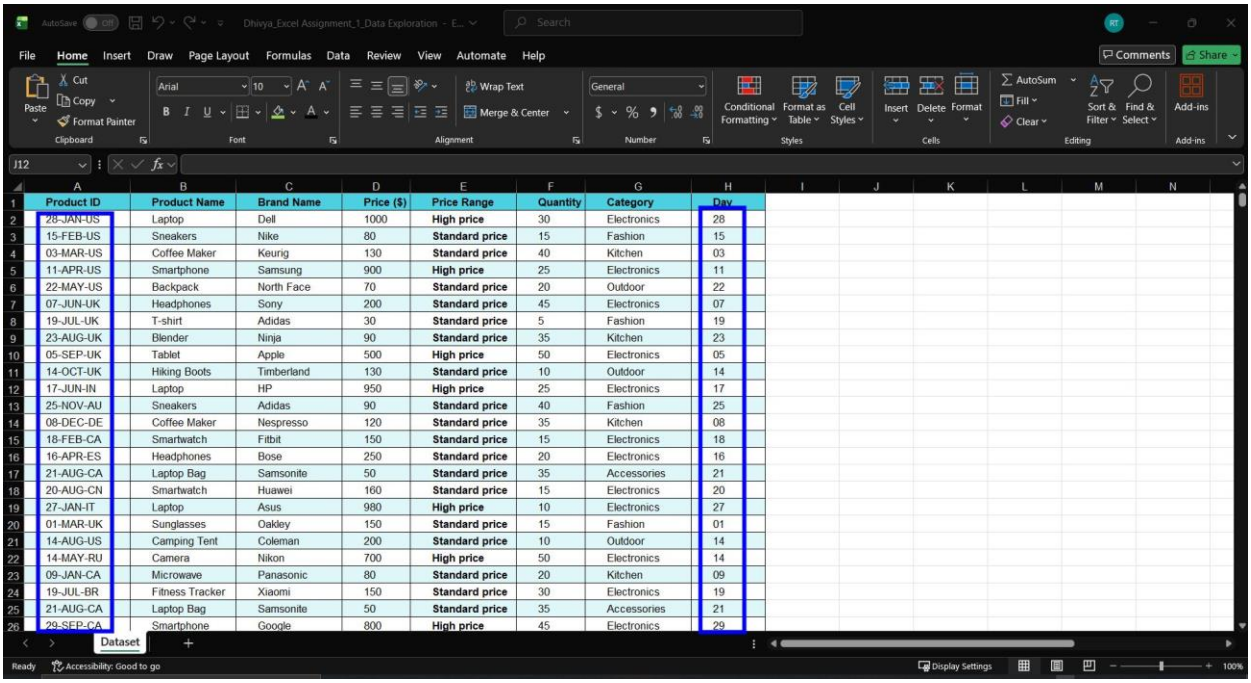


Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category
28-JAN-US	Laptop	Dell	1000	High price	30	Electronics
15-FEB-US	Sneakers	Nike	80	Standard price	15	Fashion
03-MAR-US	Coffee Maker	Keurig	130	Standard price	40	Kitchen
11-APR-US	Smartphone	Samsung	900	High price	25	Electronics
22-MAY-US	Backpack	North Face	70	Standard price	20	Outdoor
07-JUN-UK	Headphones	Sony	200	Standard price	45	Electronics
19-JUL-UK	T-shirt	Adidas	30	Standard price	5	Fashion
23-AUG-UK	Blender	Ninja	90	Standard price	35	Kitchen
05-SEP-UK	Tablet	Apple	500	High price	50	Electronics
14-OCT-UK	Hiking Boots	Timberland	130	Standard price	10	Outdoor
17-JUN-IN	Laptop	HP	950	High price	25	Electronics
25-NOV-AU	Sneakers	Adidas	90	Standard price	40	Fashion
08-DEC-DE	Coffee Maker	Nespresso	120	Standard price	35	Kitchen
18-FEB-CA	Smartwatch	Fitbit	150	Standard price	15	Electronics
16-APR-ES	Headphones	Bose	250	Standard price	20	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories
20-AUG-CN	Smartwatch	Huawei	160	Standard price	15	Electronics
27-JAN-IT	Laptop	Asus	980	High price	10	Electronics
01-MAR-UK	Sunglasses	Oakley	150	Standard price	15	Fashion
14-AUG-US	Camping Tent	Coleman	200	Standard price	10	Outdoor
14-MAY-RU	Camera	Nikon	700	High price	50	Electronics
09-JAN-CA	Microwave	Panasonic	80	Standard price	20	Kitchen
19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard price	30	Electronics
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories
29-SEP-CA	Smartphone	Google	800	High price	45	Electronics

COUNT IF	
Price less than $100	11

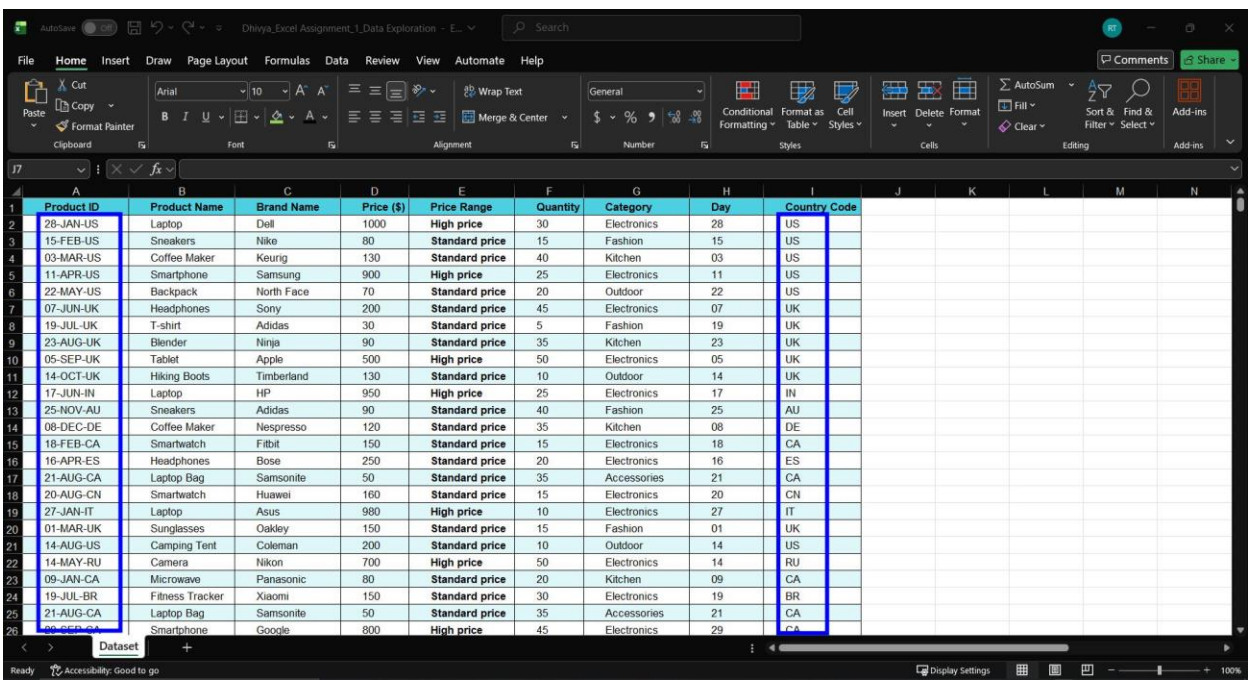
5. Text Formatting – LEFT, RIGHT, MID

LEFT FORMATTING =LEFT(A2:A35,2)



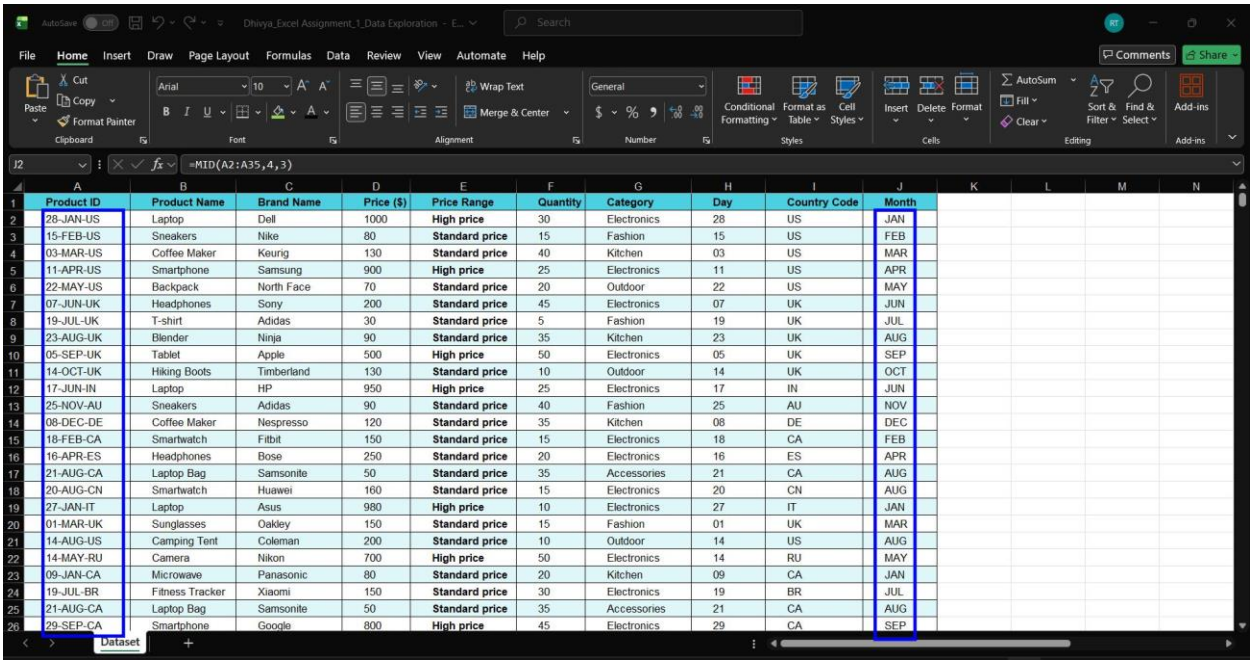
Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category	Day
28-JAN-US	Laptop	Dell	1000	High price	30	Electronics	28
15-FEB-US	Sneakers	Nike	80	Standard price	15	Fashion	15
03-MAR-US	Coffee Maker	Keurig	130	Standard price	40	Kitchen	03
11-APR-US	Smartphone	Samsung	900	High price	25	Electronics	11
22-MAY-US	Backpack	North Face	70	Standard price	20	Outdoor	22
07-JUN-UK	Headphones	Sony	200	Standard price	45	Electronics	07
19-JUL-UK	T-shirt	Adidas	30	Standard price	5	Fashion	19
23-AUG-UK	Blender	Ninja	90	Standard price	35	Kitchen	23
05-SEP-UK	Tablet	Apple	500	High price	50	Electronics	05
14-OCT-UK	Hiking Boots	Timberland	130	Standard price	10	Outdoor	14
17-JUN-IN	Laptop	HP	950	High price	25	Electronics	17
25-NOV-AU	Sneakers	Adidas	90	Standard price	40	Fashion	25
08-DEC-DE	Coffee Maker	Nespresso	120	Standard price	35	Kitchen	08
18-FEB-CA	Smartwatch	Fitbit	150	Standard price	15	Electronics	18
16-APR-ES	Headphones	Bose	250	Standard price	20	Electronics	16
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories	21
20-AUG-CN	Smartwatch	Huawei	160	Standard price	15	Electronics	20
27-JAN-IT	Laptop	Asus	980	High price	10	Electronics	27
01-MAR-UK	Sunglasses	Oakley	150	Standard price	15	Fashion	01
14-AUG-US	Camping Tent	Coleman	200	Standard price	10	Outdoor	14
14-MAY-RU	Camera	Nikon	700	High price	50	Electronics	14
09-JAN-CA	Microwave	Panasonic	80	Standard price	20	Kitchen	09
19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard price	30	Electronics	19
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories	21
29-SEP-CA	Smartphone	Google	800	High price	45	Electronics	29

RIGHT FORMATTING =RIGHT(A2:A35,2)



Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category	Day	Country Code
28-JAN-US	Laptop	Dell	1000	High price	30	Electronics	28	US
15-FEB-US	Sneakers	Nike	80	Standard price	15	Fashion	15	US
03-MAR-US	Coffee Maker	Keurig	130	Standard price	40	Kitchen	03	US
11-APR-US	Smartphone	Samsung	900	High price	25	Electronics	11	US
22-MAY-US	Backpack	North Face	70	Standard price	20	Outdoor	22	US
07-JUN-UK	Headphones	Sony	200	Standard price	45	Electronics	07	UK
19-JUL-UK	T-shirt	Adidas	30	Standard price	5	Fashion	19	UK
23-AUG-UK	Blender	Ninja	90	Standard price	35	Kitchen	23	UK
05-SEP-UK	Tablet	Apple	500	High price	50	Electronics	05	UK
14-OCT-UK	Hiking Boots	Timberland	130	Standard price	10	Outdoor	14	UK
17-JUN-IN	Laptop	HP	950	High price	25	Electronics	17	IN
25-NOV-AU	Sneakers	Adidas	90	Standard price	40	Fashion	25	AU
08-DEC-DE	Coffee Maker	Nespresso	120	Standard price	35	Kitchen	08	DE
18-FEB-CA	Smartwatch	Fitbit	150	Standard price	15	Electronics	18	CA
16-APR-ES	Headphones	Bose	250	Standard price	20	Electronics	16	ES
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories	21	CA
20-AUG-CN	Smartwatch	Huawei	160	Standard price	15	Electronics	20	CN
27-JAN-IT	Laptop	Asus	980	High price	10	Electronics	27	IT
01-MAR-UK	Sunglasses	Oakley	150	Standard price	15	Fashion	01	UK
14-AUG-US	Camping Tent	Coleman	200	Standard price	10	Outdoor	14	US
14-MAY-RU	Camera	Nikon	700	High price	50	Electronics	14	RU
09-JAN-CA	Microwave	Panasonic	80	Standard price	20	Kitchen	09	CA
19-JUL-BR	Fitness Tracker	Xiaomi	150	Standard price	30	Electronics	19	BR
21-AUG-CA	Laptop Bag	Samsonite	50	Standard price	35	Accessories	21	CA
29-SEP-CA	Smartphone	Google	800	High price	45	Electronics	29	CA

MID FORMATTING =MID(A2:A35,4,3)



J2 =MID(A2:A35,4,3)

Product ID	Product Name	Brand Name	Price ($)	Price Range	Quantity	Category	Day	Country Code	Month
28-JAN-US	Laptop	Dell	1000	**High price**	30	Electronics	28	US	JAN
15-FEB-US	Sneakers	Nike	80	**Standard price**	15	Fashion	15	US	FEB
03-MAR-US	Coffee Maker	Keurig	130	**Standard price**	40	Kitchen	03	US	MAR
11-APR-US	Smartphone	Samsung	900	**High price**	25	Electronics	11	US	APR
22-MAY-US	Backpack	North Face	70	**Standard price**	20	Outdoor	22	US	MAY
07-JUN-UK	Headphones	Sony	200	**Standard price**	45	Electronics	07	UK	JUN
19-JUL-UK	T-shirt	Adidas	30	**Standard price**	5	Fashion	19	UK	JUL
23-AUG-UK	Blender	Ninja	90	**Standard price**	35	Kitchen	23	UK	AUG
05-SEP-UK	Tablet	Apple	500	**High price**	50	Electronics	05	UK	SEP
14-OCT-UK	Hiking Boots	Timberland	130	**Standard price**	10	Outdoor	14	UK	OCT
17-JUN-IN	Laptop	HP	950	**High price**	25	Electronics	17	IN	JUN
25-NOV-AU	Sneakers	Adidas	90	**Standard price**	40	Fashion	25	AU	NOV
08-DEC-DE	Coffee Maker	Nespresso	120	**Standard price**	35	Kitchen	08	DE	DEC
18-FEB-CA	Smartwatch	Fitbit	150	**Standard price**	15	Electronics	18	CA	FEB
16-APR-ES	Headphones	Bose	250	**Standard price**	20	Electronics	16	ES	APR
21-AUG-CA	Laptop Bag	Samsonite	50	**Standard price**	35	Accessories	21	CA	AUG
20-AUG-CN	Smartwatch	Huawei	160	**Standard price**	15	Electronics	20	CN	AUG
27-JAN-IT	Laptop	Asus	980	**High price**	10	Electronics	27	IT	JAN
01-MAR-UK	Sunglasses	Oakley	150	**Standard price**	15	Fashion	01	UK	MAR
14-AUG-US	Camping Tent	Coleman	200	**Standard price**	10	Outdoor	14	US	AUG
14-MAY-RU	Camera	Nikon	700	**High price**	50	Electronics	14	RU	MAY
09-JAN-CA	Microwave	Panasonic	80	**Standard price**	20	Kitchen	09	CA	JAN
19-JUL-BR	Fitness Tracker	Xiaomi	150	**Standard price**	30	Electronics	19	BR	JUL
21-AUG-CA	Laptop Bag	Samsonite	50	**Standard price**	35	Accessories	21	CA	AUG
29-SEP-CA	Smartphone	Google	800	**High price**	45	Electronics	29	CA	SEP